

Zepto Inventory Optimization & Retail Strategy

An End-to-End SQL Analytics Report on Pricing, Categories, and Stock Health

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PostgreSQL / BI

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1. Executive Summary

This comprehensive report provides an analytical deep dive into Zepto's e-grocery inventory management, pricing strategies, and structural product breakdowns. Leveraging transactional and catalog database snapshots, the primary goal is to isolate revenue drivers, pinpoint leakage due to out-of-stock (OOS) high-value items, and provide metric-driven clarity on product discounting frameworks.

PostgreSQL

DATABASE ENGINE

8 Key Areas

STRATEGIC INSIGHTS

Optimized

INVENTORY HEALTH

By structuring raw catalog data into interactive components—ranging from price-per-gram optimization grids to cross-category distribution layers—this analysis presents actionable business frameworks to improve gross margins and prevent customer churn caused by inventory gaps.

2. Database Schema & Data Integrity

The backend engine utilizes a structured zepto table containing real-time catalog fields. Below is the Data Definition Language (DDL) pipeline designed to support fast indexing and robust queries.

```
CREATE TABLE zepto (  
  sku_id SERIAL PRIMARY KEY,  
  category VARCHAR(120),  
  name VARCHAR(150) NOT NULL,  
  mrp NUMERIC(8,2),  
  discountPercent NUMERIC(5,2),  
  availableQuantity INTEGER,  
  discountedSellingPrice NUMERIC(8,2),  
  weightInGms INTEGER,  
  outOfStock BOOLEAN,  
  quantity INTEGER  
);
```

Data Quality & Cleaning Pipelines

Before drawing statistical business inferences, data integrity checks were executed to handle anomalous entries. Zero-price records were completely removed to prevent mathematical skew in discount margins, and

price fields were transformed from underlying storage formats (paise) into standard standard currency format (rupees).

```
-- Standardizing currency representations from paise to rupees
UPDATE zepto
SET mrp = mrp / 100.0,
    discountedSellingPrice = discountedSellingPrice / 100.0;
```

3. Deep Dive Analytics & Business Insights

Insight 1: Top 10 Best-Value Products Based on Discount

Identifying steep loss-leaders helps marketing focus customer acquisition campaigns on highly appealing entry-point deals.

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
ORDER BY discountPercent DESC
LIMIT 10;
```

STRATEGIC GROWTH VIEW

Products with discounts exceeding natural thresholds should be featured prominently on app landing screens to trigger high-velocity cart adds.

Insight 2: High MRP Products Currently Out of Stock

High Maximum Retail Price (MRP) items represent significant chunks of uncaptured revenue when unavailable. This query flags items with premium price tags (> ₹300) experiencing critical inventory depletion.

```
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE AND mrp > 300
ORDER BY mrp DESC;
```

REVENUE LEAKAGE ALERT

Every hour premium items remain unstocked directly impacts Average Order Value (AOV). Supply lines for these specific SKUs require stricter SLA parameters.

Insight 3: Estimated Revenue Structure by Category

By assessing total potential revenue (Discounted Price × Available Stock), leadership can prioritize logistics support toward dominant macro-categories.

```
SELECT category,  
       SUM(discountedSellingPrice * availableQuantity) AS total_revenue  
FROM zepto  
GROUP BY category  
ORDER BY total_revenue DESC;
```

Insight 4: High Price, Low Discount Thresholds

Isolating high-value premium items (MRP > ₹500) that carry minimal discounts (< 10%) allows the marketing department to monitor organic margins on elite stock.

```
SELECT DISTINCT name, mrp, discountPercent  
FROM zepto  
WHERE mrp > 500 AND discountPercent LESS THAN 10  
ORDER BY mrp DESC;
```

Insight 5: Top 5 Categories with Highest Average Discounts

This helps monitor which store departments are relying heavily on price reductions to shift volume, possibly eroding unit economics.

```
SELECT category,  
       ROUND(AVG(discountPercent), 2) AS avg_discount  
FROM zepto  
GROUP BY category  
ORDER BY avg_discount DESC  
LIMIT 5;
```

Insight 6: Unit Economics — Price Per Gram Optimization

Evaluating the price per gram for consumer goods packed at scale ($\geq 100\text{g}$) allows us to detect which bulk options offer consumers the most competitive value.

```
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
                ROUND(discountedSellingPrice / weightInGms, 2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram ASC;
```

Insight 7: Weight Matrix Breakdown (Low vs Medium vs Bulk)

Segmenting the product catalog dynamically into Low ($<1\text{kg}$), Medium ($<5\text{kg}$), and Bulk ($\geq 5\text{kg}$) categories creates structured fulfillment routing parameters for warehouse logistics.

```
SELECT DISTINCT name, weightInGms,
                CASE WHEN weightInGms < 1000 THEN 'Low'
                     WHEN weightInGms < 5000 THEN 'Medium'
                     ELSE 'Bulk'
                END AS weight_category
FROM zepto;
```

Insight 8: Total Inventory Weight Burden per Category

Calculating total volumetric kilograms per catalog tier guides truck loading logistics and dark-store shelving weight constraints.

```
SELECT category,
        SUM(weightInGms * availableQuantity) AS total_weight
FROM zepto
GROUP BY category
ORDER BY total_weight DESC;
```

4. Data-Driven Recommendations

1. **Automated Stock Triggers:** Implement automated replenishment webhooks for any items falling under Insight 2 (High MRP, Out of Stock) to avoid high-margin revenue losses.

2. **Volumetric Allocation:** Cross-reference Insight 8's weight burden figures with active storage dimensions at dark store hubs to optimize physical floor footprint utilization.

3. **Dynamic Pricing Safeguards:** Cap category discounts within Insight 5 if profit margins slip past predefined operational corridors.